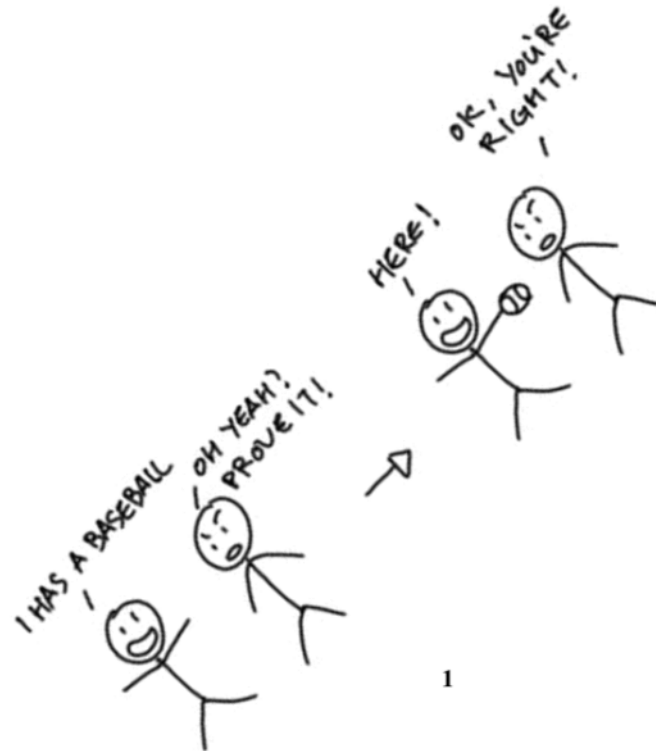

COMP 2121
DISCRETE
MATHEMATICS

Lecture 6



Logical Implication: Rules of Inference

¹ <http://img522.imageshack.us/img522/4043/clipboard3781kr2.jpg>

Valid Argument

➤ Logical Implication: Rules of Inference

In mathematics, an argument is defined to be a sequence of statements that reach a conclusion. If the argument is *valid*, then the conclusion follows from the earlier statements in the argument. All of the statements, except the final one are called premises (or assumptions, or hypotheses). The final statement is called the conclusion. We often use the symbol $\therefore$ (read “therefore”) just before the conclusion.

An argument form is an argument involving statement variables (such as p and q). An argument form is *valid* when any substitution of statements for the statement variables that leaves the premises true also results in the conclusion being true. To say an argument is valid means that its form is valid.

Example 1

Example 1. The following is the classic argument:

If ^pSocrates is a human being, then ^qSocrates is mortal;
Socrates is a human being;
 $\therefore$ Socrates is mortal.

This argument has the abstract form:

premises { if p then q ;
conclusion { p ;
 $\therefore q$.

In general, we can have n premises, $p_1, p_2, \dots, p_n$. The argument $(p_1 \wedge p_2 \wedge p_3 \cdots \wedge p_n) \rightarrow q$ is called valid if whenever each of premises $p_1, p_2, \dots, p_n$ is true, the conclusion q is likewise true.

Testing validity using Truth Table

Therefore, we can test an argument form for validity by following these steps:

1. Identify the premises and conclusion of the argument.
2. Construct a truth table showing the truth values of all the premises and the conclusion.
3. Find the rows (*critical rows*) for which all of the premises are true.
4. For each critical row, check if the conclusion is also true.
 - If the conclusion is true for all critical rows, the argument form is valid.
 - If one or more critical rows have a false conclusion, then the argument form is invalid.

Example 2. Show that the following argument form is valid:

$$\left. \begin{array}{l} p \vee (q \vee r) \\ \neg r \end{array} \right\} \therefore p \vee q.$$

p	q	r	$q \vee r$	$p \vee (q \vee r)$	$\neg r$	$p \vee q$
0	0	0	0	0	1	0
0	0	1	1	1	0	0
0	1	0	1	1	1	1
0	1	1	1	1	0	1
1	0	0	0	1	1	1
1	0	1	1	1	0	1
1	1	0	1	1	1	1
1	1	1	1	1	0	1

Modus Ponens/Modus Tollens

Modus Ponens – Rule of Detachment. There are several standard argument forms used. See the text for several additional examples. The one we saw above, with Socrates, is called modus ponens. It is extremely common:

$$\begin{array}{l} \text{if } p \text{ then } q; \\ p; \\ \therefore q. \end{array}$$

p	q	$p \rightarrow q$	p	q
0	0	1	0	0
0	1	1	0	1
1	0	0	1	0
1	1	1	1	1

$$\begin{array}{l} a \rightarrow b \\ a \\ \hline \therefore b \end{array}$$

Modus Tollens - Method of Denying. A related form is modus tollens, which takes the form:

given $\left\{ \begin{array}{l} \text{if } p \text{ then } q; \\ \neg q; \end{array} \right.$
 $\therefore \neg p.$

1) $p \rightarrow q$
 2) $\neg q$
 3) $\neg q \rightarrow \neg p$
 4) $\neg p$

given
 given
 contraposition 1
 Modus Ponens 3, 4

An example of this is

1. If Zeus is human, then Zeus is mortal
2. Zeus is not mortal
3. Therefore, Zeus is not human. (Identify p and q here).

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$
} M.T.

Rules Of Inference

$$\begin{array}{l}
 \frac{p}{p \rightarrow q} \\
 \hline
 \therefore q
 \end{array}
 \quad
 \frac{a}{a \rightarrow b} \\
 \hline
 \therefore b
 \quad
 \frac{a \rightarrow b}{a} \\
 \hline
 \therefore b
 \quad
 \frac{(7a \wedge b) \rightarrow 7c}{7a \wedge b} \\
 \hline
 \therefore 7c$$

c.s. $\frac{a \wedge b}{\therefore a}$

$$\frac{(a \wedge b) \rightarrow x}{\therefore a \rightarrow x} !!!$$

Rule of Inference	Related Logical Implication	Name of Rule
1) $\frac{p}{p \rightarrow q}$ $\hline \therefore q$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Rule of Detachment (Modus Ponens)
2) $\frac{p \rightarrow q}{q \rightarrow r}$ $\hline \therefore p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Law of the Syllogism
3) $\frac{p \rightarrow q}{\neg q}$ $\hline \therefore \neg p$	$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$	Modus Tollens
4) $\frac{p}{q}$ $\hline \therefore p \wedge q$		Rule of Conjunction
5) $\frac{p \vee q}{\neg p}$ $\hline \therefore q$	$[(p \vee q) \wedge \neg p] \rightarrow q$	Rule of Disjunctive Syllogism
6) $\frac{\neg p \rightarrow F_0}{\therefore p}$	$(\neg p \rightarrow F_0) \rightarrow p$	Rule of Contradiction
7) $\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Rule of Conjunctive Simplification
8) $\frac{p}{\therefore p \vee q}$	$p \rightarrow p \vee q$	Rule of Disjunctive Amplification
9) $\frac{p \wedge q}{p \rightarrow (q \rightarrow r)}$ $\hline \therefore r$	$[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$	Rule of Conditional Proof
10) $\frac{p \rightarrow r}{q \rightarrow r}$ $\hline \therefore (p \vee q) \rightarrow r$	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow (p \vee q) \rightarrow r$	Rule for Proof by Cases
11) $\frac{p \rightarrow q}{r \rightarrow s}$ $\hline \frac{p \vee r}{\therefore q \vee s}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$	Rule of the Constructive Dilemma
12) $\frac{p \rightarrow q}{r \rightarrow s}$ $\hline \frac{\neg q \vee \neg s}{\therefore \neg p \vee \neg r}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$	Rule of the Destructive Dilemma

Example 3

Example 3. Show that the following argument is valid using our known argument forms:

Given
 $p \rightarrow x$
 $p \rightarrow (w \wedge x)$
 $\neg q$
 $p \vee q$
 $\therefore w$

$a \rightarrow b$
 a
 $\therefore b$

	STEP	REASON
1)	$\neg q$	given
2)	$p \vee q$	given
3)	p	D.S. 1, 2

4)	$p \rightarrow x$	given
5)	x	M.P. 3, 4

6)	$p \rightarrow (w \wedge x)$	given
7)	$w \wedge x$	M.P. 3, 6
8)	w	C.S. 7

→ given
 → rule of logic
 → Rule of inference

Example 4

Example 4. Prove that the following argument is valid:

$$\neg p \rightarrow (r \wedge \neg s) \checkmark$$

$$x \rightarrow s$$

$$u \rightarrow \neg p \checkmark$$

$$\neg w \checkmark$$

$$u \vee w \checkmark$$

$$\therefore \neg x \vee w$$

$$\begin{array}{l} a \rightarrow b \\ a \\ \hline \therefore b \end{array} \quad \text{H.P.}$$

M.T.

$$a \rightarrow b$$

$$\neg b$$

$$\therefore \neg a$$

$$\frac{a}{a \vee b}$$

STEPS

$$1) \neg w$$

$$2) u \vee w$$

$$3) u$$

$$4) u \rightarrow \neg p$$

$$5) \neg p$$

$$6) \neg p \rightarrow (r \wedge \neg s)$$

$$7) r \wedge \neg s$$

$$8) r$$

$$9) \neg s$$

$$10) x \rightarrow s$$

11) $\neg x$	M.T. 9, 10
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$$11) \neg s \rightarrow \neg x$$

$$12) \neg x$$

$$13) \neg x \vee w$$

Reasons

given

given

D.S. 1, 2

given

H.P. 3, 4

given

H.P. 5, 6

C.S. 7

C.S. 7

given

Contrapos. 10

M.P. 9, 11

D.A. 12